

Primewave AppV1

Why Home feels slow — every function until the page is shown

From tapping Login (or opening the app) through Lottie → skeleton → real Home. Each function is listed in time order with what it is for and how expensive it is.

2026-08-17 · dashboard-v2 Home tab

Load is “done” when `revealStep = 5` (header, badges, scenes, Home Access, rooms are real) **and** `frigateConfigResolved` (camera strip leaves skeleton). Until then you only see shimmer.

0. Why the skeleton lasts so long (root causes)

The skeleton is not a decoration — Home is hidden on purpose until three things finish. Anything slower than those three keeps you on shimmer.

- **Hard wait (reveal gate):** HA `get_states` must return *and* `GET /api/quick-scenes` *and* `GET /api/home-access`. The slowest of these three wins.
- **Lottie (`app/index.jsx`):** forced **2.2 seconds** even if SecureStore is instant. This runs on **app open** (route `/`), not inside `dashboard-v2`. After you tap Login, code goes **directly** to `/dashboard-v2` — no second Lottie. If you see Lottie then skeleton, that Lottie was the splash before/during first navigation.
- **`get_states` is huge:** Home Assistant sends *every* entity (often 500–2000). Parsing that JSON on the phone is usually the longest wait.
- **Then JS work on the main thread:** `getRoomsWithCounts` → `getRoomEntities` for every room does `.find()` over the full entity list. That can freeze the UI after data arrives (skeleton stays, then a hitch, then the page pops in).
- **Extra parallel traffic (does not block reveal, but steals bandwidth/CPU):** ~15 admin APIs, 7 more HA registry calls, Frigate config, notification history, heartbeat, light-icon SVG preload. Sensors and covers are fetched **twice**.
- **Cameras after reveal:** each `HomeCameraStrip` card starts a `WebView` stream. That can make the page feel unfinished even after rooms appear.

SLOW = likely seconds on a real phone **MED** = tens–hundreds of ms or extra network **fast** = local / tiny. Red rows are the ones to attack first.

1. App open — Lottie splash (`app/index.jsx`) · always on cold start

Expo Router first screen is `index`. You see `PrimeWave2.json` until session check **and** 2200 ms have both finished. Then replace → `login` (no session) or `dashboard-v2` (session).

#	Function	File	What it does (until Home is shown)	Cost
1	<code>SplashScreen.preventAutoHideAsync</code>	<code>_layout.jsx</code>	Keeps native splash up until fonts load.	fast
2	<code>useFonts (Clash Display × 6)</code>	<code>_layout.jsx</code>	Blocks first JS frame until 6 OTF files are in memory.	MED
3	<code>preloadLocalLightIcons</code>	<code>lightTypeAssets.js</code>	Reads 6 light SVGs from disk, tints white + black. Started here, not required for Home cards.	MED
4	<code>registerForPushNotificationsAsync</code>	<code>notifications.js</code>	Permission, Expo token, <code>POST /api/notifications/register</code> . On first install <code>adminUrl</code> may be missing so this no-ops until login.	MED
5	<code>ScreenOrientation.lockAsync</code>	<code>_layout.jsx</code>	Portrait lock on phones.	fast
6	<code>getLastNotificationResponseAsync</code>	<code>_layout.jsx</code>	If launched from a push, stash it in <code>NotifContext</code> .	fast
7	<code>LottieView PrimeWave2.json</code>	<code>index.jsx</code>	Plays branded animation. Hides native splash on layout.	MED
8	<code>setTimeout 2200 ms</code>	<code>index.jsx</code> <code>MIN_SPLASH_MS</code>	Forces you to watch Lottie at least 2.2s even if login check is done.	SLOW
9	<code>SecureStore.getItemAsync × 4</code>	<code>index.jsx</code>	<code>is_logged_in</code> , <code>ha_active_profile_id</code> , <code>ha_profiles</code> , <code>logged_in_user</code> — decide login vs dashboard.	fast
10	<code>router.replace</code>	<code>index.jsx</code>	→ <code>/login</code> or <code>/dashboard-v2</code> with <code>userName</code> , <code>userId</code> .	fast

2. Tap Login (`app/login.jsx handleLogin`) · only if you are on the login screen

After success this does `router.replace('/dashboard-v2')`. It does **not** play Lottie again.

#	Function	File	What it does (until Home is shown)	Cost
11	<code>handleLogin</code>	<code>login.jsx</code>	Validates fields, sets <code>isLoggingIn</code> spinner on the button.	fast
12	<code>validateCredentials</code>	<code>services/auth.js</code>	Two sequential HTTP calls to HA: <code>POST /auth/login_flow</code> then <code>POST /auth/login_flow/{id}</code> with <code>username/password</code> . Network RTT × 2.	SLOW
13	<code>SecureStore get profiles</code>	<code>login.jsx</code>	Read active profile name for the account record.	fast

#	Function	File	What it does (until Home is shown)	Cost
14	upsertAccountAndActivate	services/accounts.js	Writes saved_accounts, active_account_id, is_logged_in, logged_in_user, Face ID username/password.	MED
15	activateAccount	accounts.js	Sets ha_active_profile_id + session flags (called from upsert).	fast
16	registerForPushNotificationsAsync	notifications.js	Fired again now that adminUrl exists (layout ran too early on fresh install). GET token + POST register. Does not block navigation.	MED
17	getAdminUrl	utils/storage.js	Used inside push register to find backend URL.	fast
18	router.replace /dashboard-v2	login.jsx	Unmounts login, mounts DashboardV2. Skeleton period starts here.	fast

3. Dashboard mounts — you now see SKELETONS (this is the long wait)

DashboardV2 first render: connectionConfig.loaded is false → full DashboardSkeleton. Then profile loads → per-section skeletons (header/scenes/access/rooms/cameras) until the reveal gate.

3.1 Local profile + settings (parallel, first tick)

#	Function	File	What it does (until Home is shown)	Cost
19	loadConnectionConfig	dashboard-v2.jsx	Reads ha_active_profile_id + ha_profiles. Fills url, token, adminUrl, loaded=true. Unlocks all network.	fast
20	SecureStore settings × 6	dashboard-v2.jsx mount effect	show family, auto room visit/resume, voice assistant, preference button, room_reorder_config.	fast
21	getButlerBackendUrl	butlerBackend.js	Builds adminUrl/api/butler. No socket. Not needed for Home paint.	fast
22	useDeviceType	hooks/useDeviceType.js	Phone vs tablet column counts for skeletons and grids.	fast
23	useNotifications	hooks/useNotifications.js	Starts fetch of history+log once adminUrl exists (see 48–49).	fast
24	useHaSystemHealth	hooks/useHaSystemHealth.js	Watches haStatus/adminStatus; banner only after 10s of failure.	fast

3.2 Home Assistant WebSocket — usually the #1 delay

#	Function	File	What it does (until Home is shown)	Cost
25	new HAService(url, token)	services/ha.js	Builds wss://.../api/websocket. Subscribes to AppState for later reconnect.	fast
26	HAService.connect	ha.js	Opens WebSocket. Waits for TCP + TLS to HA. Slow on remote / weak Wi-Fi.	SLOW
27	sendAuth	ha.js	Sends access_token when HA says auth_required.	fast
28	handleMessage auth_ok	ha.js	Marks connected, notifies dashboard, then subscribe_events.	fast
29	subscribe_events state_changed	ha.js	Live stream for the rest of the session. Not required to leave skeleton.	fast
30	getStates	ha.js	HA command get_states. Returns ALL entities. Largest payload. Dashboard will not reveal without this.	SLOW
31	getConfig	ha.js	HA location_name → header city. In the same Promise.all as get_states.	MED
32	getDeviceRegistry	ha.js	config/device_registry/list — devices → rooms.	MED
33	getEntityRegistry	ha.js	config/entity_registry/list — often as large as get_states.	SLOW
34	getCategoryRegistry	ha.js	Loaded in the same batch. Not drawn on Home.	MED
35	getAreaRegistry	ha.js	Rooms list. Required for room cards.	MED
36	getFloorRegistry	ha.js	Floors. Home strip does not use floors; Rooms tab does.	fast

#	Function	File	What it does (until Home is shown)	Cost
37	getConfigEntries	ha.js	Music Assistant entry ids. Failure ignored. Not needed for Home first paint.	MED
38	Promise.all then setState × 8	dashboard-v2.jsx	One React batch: entities, city, devices, registries, floors, MA ids. Triggers the expensive room memos.	SLOW

3.3 Admin APIs — all fire together (fetchMappings + initial-load)

Two useEffects both hit the admin server. **Bold = reveal gate**. Sensors and covers are requested in both effects (duplicate).

#	Function	File	What it does (until Home is shown)	Cost
39	fetchMappings	dashboard-v2.jsx	Aborts previous, starts the mapping fan-out. Resets reveal to 0 on first load.	fast
40	GET /api/quick-scenes	fetchMappings	IDs for the 4 scene cards. Sets scenesFetched. GATE.	SLOW
41	fetchEnrichedLightMappings	lightMappingsClient.js	Parallel GET monitored-entities?type=light AND GET /api/light-types, then enrichLightMappings.	MED
42	enrichLightMappings / inferLightTypeFromEntity	lightTypeInference.js	Guess icon type from entity_id keywords for every mapped light.	MED
43	preloadLocalLightIcons	lightTypeAssets.js	Called again after light-types. Disk SVG + tintSvgXml.	MED
44	GET /api/monitored-entities?type=media_player	fetchMappings	TV mappings. Not needed to show Home rooms.	MED
45	GET /api/sensors	fetchMappings + initial-load DUPLICATE	Door types for room badges. Fetched twice.	MED
46	GET /api/covers	fetchMappings + initial-load DUPLICATE	Garage types + windows. Fetched twice.	MED
47	GET /api/climate-mappings	fetchMappings	AC dampers. Not on Home strip.	MED
48	GET /api/home-access	fetchMappings	Which locks + garages. Sets homeAccessFetched. GATE.	SLOW
49	GET /api/lock-passages	fetchMappings	Lock ↔ door sensor for StatusBadges dots.	MED
50	GET /api/config	initial-load effect	selected_areas, selected_cameras, locks_armed, badge entities. Needed for which rooms to show.	SLOW
51	GET /api/alerts	initial-load	PersonBadges alert chips.	MED
52	GET /api/entities	initial-load	Monitored/ignored sets for notifications. Not drawn.	MED
53	GET /api/room-tracking/lookup	initial-load	Presence → room. May open a sheet after Home is up.	MED
54	GET /api/notifications/history	useNotifications.fetchNotifications	Bell unread. Promise.allSettled with log.	MED
55	GET /api/notifications/log	useNotifications	Lock/garage push log for the bell.	MED
56	startHeartbeat → POST /api/sessions/heartbeat	heartbeat.js	Immediately then every 30s. Device id + user. Invisible.	MED
57	getloIdForVendorAsync / getAndroidId	heartbeat.js getDeviceId	Used inside heartbeat.	fast

3.4 Frigate / cameras (strip skeleton only)

#	Function	File	What it does (until Home is shown)	Cost
58	new FrigateService	services/frigate.js	Points at admin proxy, not a raw Frigate IP.	fast
59	FrigateService.getConfig	frigate.js	GET {admin}/api/frigate/config. Sets cameras. On fail may Alert. Always sets frigateConfigResolved.	SLOW
60	GET /api/cameras	dashboard-v2.jsx	HA camera fallback if Frigate list is empty.	MED

#	Function	File	What it does (until Home is shown)	Cost
61	getStreamUrl / getHASnapshotUrl	frigate.js	Called when HomeCameraStrip mounts — one WebView per visible camera.	SLOW

4. After data arrives — JS that builds the Home you see (can hitch the skeleton)

When Promise.all and the two GATE APIs finish, React re-renders. These functions run on the UI thread. If get_states was large, this is the freeze right before the page appears.

#	Function	File	What it does (until Home is shown)	Cost
62	getSelectedAreasForDashboard	utils/roomAreas.js	Intersect HA areas with /api/config selected_areas.	fast
63	filterParentRoomsForDashboard	roomAreas.js	Hide sub-rooms (Toilet, etc.) from the Home strip.	fast
64	getRoomAreaGroup	roomAreas.js	Merge child-area stats onto parent room cards.	fast
65	getRoomsWithCounts	dashboard-v2.jsx	For every parent area, compute badges. Calls getRoomEntities per room. Main CPU cost.	SLOW
66	getRoomEntities	utils/roomHelpers.js	Filter registry by area, then .find() live state for each entity. Nested loops over all entities × rooms.	SLOW
67	inferSensorType / isRoomClimateSensor	roomHelpers.js	Door vs temp vs humidity for badges.	fast
68	findCoverMapping / inferCoverLayer / isMasterCover	coverWindows.js	Garage vs curtain vs shutter while building rooms.	fast
69	isMusicAssistantMediaPlayer / findLinkedRemote	roomHelpers / tvRemote	Split TV vs speakers. Extra .find on entities.	MED
70	collectGroupedLightMembers	lightCapabilities.js	HA light groups so badge counts don't double-count.	MED
71	isLightCountableUnit / countActiveCountableLights	lightCapabilities.js	Lights-on number in StatusBadges = sum of room badges.	MED
72	isMasterControllerLight / isLightGroupEntity	lightCapabilities.js	Skip master controller in counts.	fast
73	getAllActiveDevices('ac')	dashboard-v2.jsx	AC-on chip. Walks areas × registry × entities.	MED
74	homeLocks / homeCovers memos	dashboard-v2.jsx	Filter locks by home-access IDs; garages from coverMappings + live state.	MED
75	quickScenesData memo	dashboard-v2.jsx	Match scene IDs to live entities, max 4.	fast
76	weather / humidity / indoorTemp memos	dashboard-v2.jsx	Header climate line. .find on entities.	fast
77	power / securityState memo	dashboard-v2.jsx	From badgeConfig entity ids.	fast
78	buildEntityMap	CameraSensorOverlay.js	Map of all entities for camera overlays.	MED
79	analyzeEntitiesHealth	haEntityHealth.js	Inside useHaSystemHealth — % unavailable.	fast
80	HeaderV2 render	HeaderV2.jsx	Greeting, weather icon, temps, bell, mic.	fast
81	StatusBadges render	StatusBadges.jsx	Lock dots + lights + AC chips.	fast
82	PersonBadges render	PersonBadges.jsx	If showFamily. person.* pictures from HA /local/...	MED
83	QuickScenes render	QuickScenes.jsx	Four SceneCards.	fast
84	HomeAccess render	HomeAccess.jsx	Lock + garage pills (Reanimated).	MED
85	RoomsList render	RoomsList.jsx	Room photos via expo-image (HA authenticated URLs).	SLOW
86	HomeCameraStrip + CameraCard WebView	HomeCameraStrip.jsx	N live streams. Heavy after reveal.	SLOW
87	TabBar / TabletSidebar	TabBar.jsx	Always visible under Home.	fast

#	Function	File	What it does (until Home is shown)	Cost
88	HaSystemBanner	HaSystemBanner.jsx	Usually nothing.	fast

5. After Home appears — still running, not required to leave skeleton

#	Function	File	What it does (until Home is shown)	Cost
89	HAService.subscribe state_changed	dashboard-v2.jsx	applyHaStateChangedEvent on every HA change. Keeps Home live.	MED
90	applyHaStateChangedEvent / applyClimateServiceToEntity	haEntityMerge.js	Patch one entity in the array.	fast
91	pushNotification	dashboard-v2.jsx	In-app list for monitored entities (locks/garage skipped — backend only).	fast
92	navigateToPresenceRoom	dashboard-v2.jsx	May open RoomSheet if tracker matches. Feels like extra load.	MED
93	AppState listener	dashboard-v2.jsx	On resume: fetchMappings again (no skeleton reset) + presence.	MED
94	canOpenButlerCall / runButlerBackgroundSetup	openButlerCall.js	NOT called until mic tap. Not part of Home load.	fast
95	callService	dashboard-v2.jsx	Only when you tap a scene/lock/garage. Optimistic state for on/off.	fast

6. Exact “page load ended” condition

- connectionConfig.loaded === true (profile from SecureStore).
- entities.length > 0 ← function **getStates** (#30).
- scenesFetched === true ← GET /api/quick-scenes (#40), even if empty or error.
- homeAccessFetched === true ← GET /api/home-access (#48), even if error.
- Then revealStep jumps 0 → 5: Header, StatusBadges, QuickScenes, HomeAccess, RoomsList swap from skeleton to real.
- Camera strip: extra wait on frigateConfigResolved (#59). Then WebViews (#86).
- Lottie 2.2s (#8) is **before** this if you came from app open via index.jsx.

Fastest wins if you later optimize: (1) don't wait 2.2s on Lottie, (2) don't block Home on home-access/quick-scenes — show rooms from HA registries first, (3) shrink get_states or index entities by id so getRoomEntities is O(1) not O(n²), (4) stop duplicate /api/sensors and /api/covers, (5) defer camera WebViews until after first paint.